2025 TRIAL

SEQUOIA — EDUCATION —

Write your **full name** in the space above.

Specialist Mathematics Examination 2

Question and Answer Book

VCE Trial Examination – 2025

- Reading time is **15 minutes**
- Writing time is **2 hours**

Approved materials

- Protractors, set squares and aids for curve sketching
- One bound reference
- One approved CAS calculator or CAS software, and one scientific calculator

Materials supplied

- Question and Answer Book of 24 pages
- Formula Sheet
- Multiple-Choice Answer Sheet

Instructions

- Follow the instructions on your Multiple-Choice Answer Sheet.
- At the end of the examination, place your Multiple-Choice Answer Sheet inside the front cover of this book.

Students are **not** permitted to bring mobile phones and/or any unauthorised electronic devices into the examination room.

Contents pages

Section A (20 questions, 20 marks) 2–9
Section B (6 questions, 60 marks) 10–21

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Section A – Multiple-choice questions

Instructions

- Answer **all** questions in pencil on your Multiple-Choice Answer Sheet.
- Choose the response that is **correct** for the question.
- A correct answer scores 1; an incorrect answer scores 0.
- Marks will **not** be deducted for incorrect answers.
- No marks will be given if more than one answer is completed for any question.
- Unless otherwise indicated, the diagrams in this book are **not** drawn to scale.
- Take the **acceleration due to gravity** to have magnitude g m s^{-2}, where $g = 9.8$

Question 1

Consider the following program. The modulus operator `%` used in line 5 returns the remainder when one integer is divided by another. For example, `10 % 3` returns `1` because when 10 is divided by 3, the remainder is 1.

```
1   n ← 6
2   count ← 0
3
4   while n > 1 do
5       if n % 2 = 0 then
6           n ← n / 2
7       else
8           n ← 3 * n + 1
9       end if
10      count ← count + 1
11  end while
```

The final value of `count` after the program terminates is

A. 5

B. 6

C. 7

D. 8

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Question 2

Consider the following statement.

'All musicians in the band can read sheet music.'

Which one of the following options is the negation of this statement?

A. No musicians in the band can read sheet music.

B. Some musicians in the band can read sheet music.

C. Some musicians in the band cannot read sheet music.

D. All musicians in the band cannot read sheet music.

Question 3

The maximal domain of the function $g(x) = \arccos\big(2 - |x+1|\big)$ is

A. $[-2,0] \cup [2,4]$

B. $[-4,-2] \cup [0,2]$

C. $[-3,1]$

D. $[-1,1]$

Question 4

Consider the hyperbola given by $4x^2 = y^2 - 2y$.

The equations of its asymptotes are

A. $y = 2x+1$ and $y = -2x+1$

B. $y = 2x-1$ and $y = -2x-1$

C. $y = 4x+1$ and $y = -4x+1$

D. $y = 4x-1$ and $y = -4x-1$

Question 5

Consider the point representing $z \in \mathbb{C} \setminus \mathbb{R}$ on an Argand diagram.

The point representing $-\bar{z}$ can be located by

A. rotating the point representing z clockwise about the origin by $90°$.

B. rotating the point representing z anticlockwise about the origin by $90°$.

C. reflecting the point representing z in the real axis.

D. reflecting the point representing z in the imaginary axis.

Question 6

Let z and w be complex numbers such that $\text{Arg}(z) = \dfrac{\pi}{6}$ and $\text{Arg}(w) = -\dfrac{\pi}{3}$.

Then $\text{Arg}\left(\dfrac{z^3}{w^2}\right)$ is equal to

A. $-\dfrac{\pi}{6}$

B. $-\dfrac{5\pi}{6}$

C. $\dfrac{5\pi}{6}$

D. $\dfrac{7\pi}{6}$

Question 7

Consider the curve defined by the parametric equations

$$x = 2\cot(t) + 1, \quad y = \csc(t) - 3, \quad 0 < t < \frac{\pi}{2}.$$

Then $\dfrac{dy}{dx}$ is equal to

A. $2\sec(t)$

B. $\dfrac{1}{2}\sec(t)$

C. $2\cos(t)$

D. $\dfrac{1}{2}\cos(t)$

Question 8

Using a suitable substitution, $\displaystyle\int_0^1 \frac{x}{\sqrt{3-x}}\,dx$ can be expressed as

A. $\displaystyle\int_2^3 \left(\frac{3}{\sqrt{u}} - \sqrt{u}\right) du$

B. $\displaystyle\int_2^3 \left(\sqrt{u} - \frac{3}{\sqrt{u}}\right) du$

C. $\displaystyle\int_0^1 \left(\frac{3}{\sqrt{u}} - \sqrt{u}\right) du$

D. $\displaystyle\int_0^1 \left(\sqrt{u} - \frac{3}{\sqrt{u}}\right) du$

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Question 9

Let $f : \mathbb{R} \to \mathbb{R}$ be a strictly convex (strictly concave up) function.

Which one of the following statements is necessarily true?

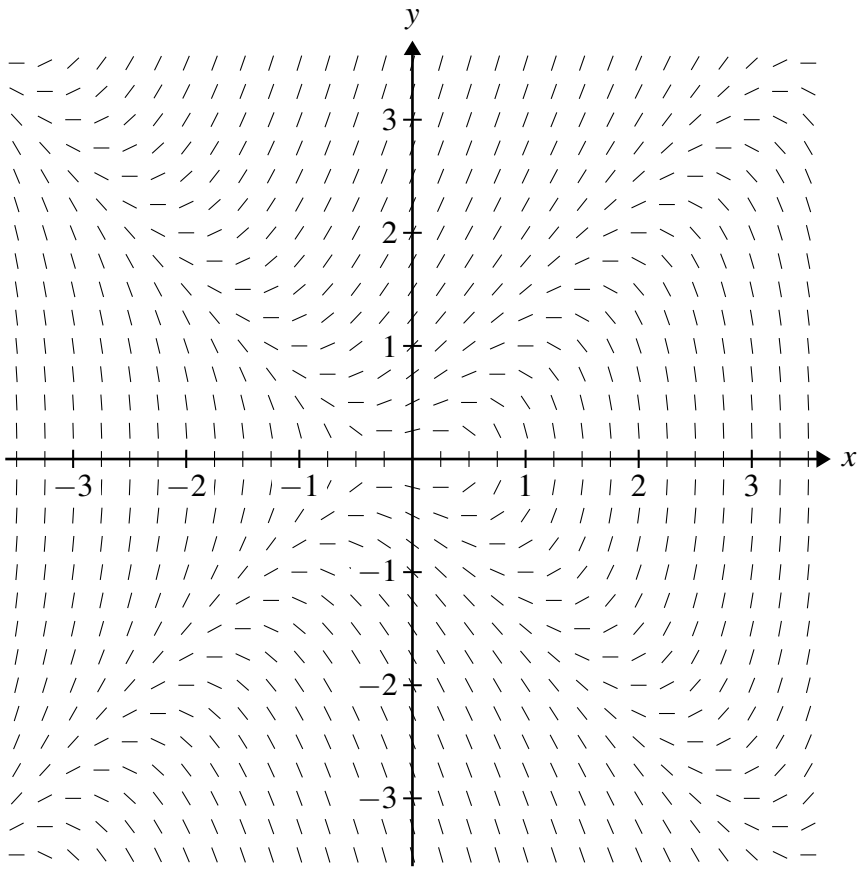
A. The function f attains a global minimum.

B. If f is differentiable then f' is strictly increasing.

C. If f is twice differentiable then $f''(x) > 0$ for all $x \in \mathbb{R}$.

D. For all $k \in \mathbb{R}$, the function kf is strictly convex.

Question 10



The slope field above best represents the differential equation

A. $\dfrac{dy}{dx} = \dfrac{x^2 - y^2}{x}$

B. $\dfrac{dy}{dx} = \dfrac{x^2 - y^2}{y}$

C. $\dfrac{dy}{dx} = \dfrac{y^2 - x^2}{x}$

D. $\dfrac{dy}{dx} = \dfrac{y^2 - x^2}{y}$

Question 11

Consider the differential equation $\dfrac{dy}{dx} = x^2 + y^2$, where $y(0) = y_0 = 1$.

Using Euler's method with a step size of 0.25, an approximation of $y(-0.5)$ is

A. 0.59375

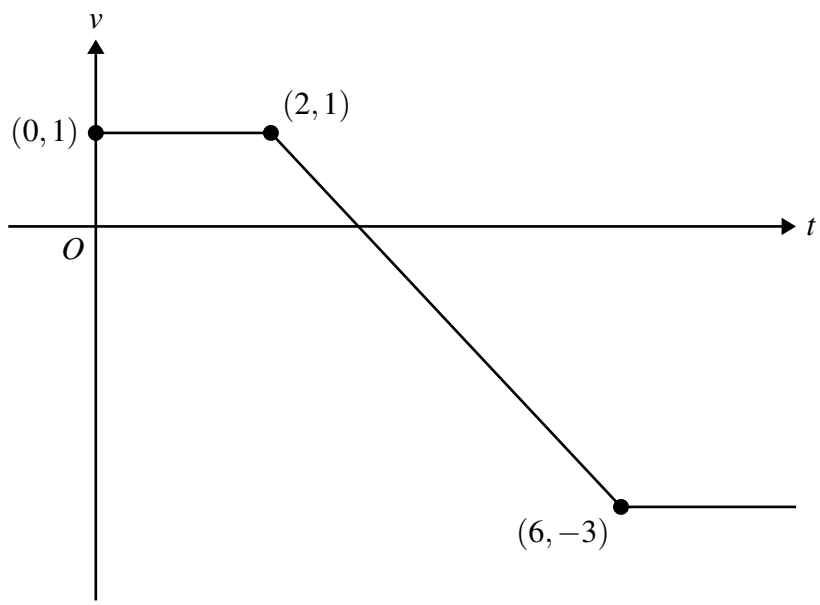
B. 0.75

C. 1.25

D. 1.65625

Question 12

The graph below shows the velocity, v m s^{-1}, of a particle moving along a straight line after t seconds. The graph of v consists only of straight-line segments.



The time, in seconds, when the particle returns to its initial position is

A. 3

B. $\sqrt{5}+3$

C. 5

D. $2\sqrt{3}+2$

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Question 13

A helicopter is hovering high above the ground. A skydiver falls from the helicopter from rest, experiencing a downward acceleration, in m s^{-2}, of $a = g - 0.01v^2$, where v is the speed of the skydiver in m s^{-1}.

The distance, in metres, through which the skydivers falls to reach a speed of 30 m s^{-1} is given by

A. $\int_0^{30} \left(g - 0.01v^2\right) dv$

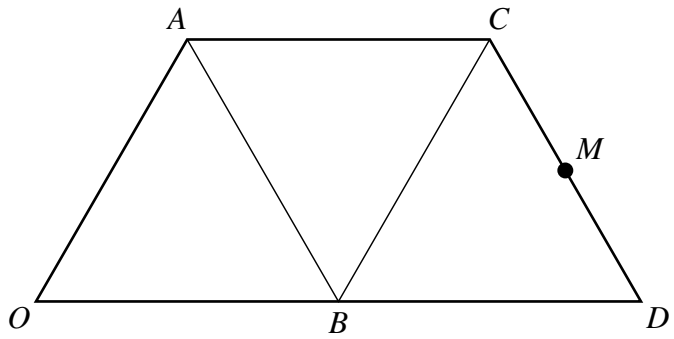
B. $\int_0^{30} \frac{g - 0.01v^2}{v} dv$

C. $\int_0^{30} \frac{1}{g - 0.01v^2} dv$

D. $\int_0^{30} \frac{v}{g - 0.01v^2} dv$

Question 14

Let $OACD$ be an isosceles trapezium constructed from three equilateral triangles, as shown below.



Let $\overrightarrow{OA} = \underset{\sim}{a}$ and $\overrightarrow{OB} = \underset{\sim}{b}$, and let M be the midpoint of CD.

The vector $\overrightarrow{OM}$ is equal to

A. $\frac{1}{2}\underset{\sim}{a} + \frac{1}{2}\underset{\sim}{b}$

B. $\frac{3}{2}\underset{\sim}{a} + \frac{1}{2}\underset{\sim}{b}$

C. $\frac{1}{2}\underset{\sim}{a} + \frac{3}{2}\underset{\sim}{b}$

D. $\frac{3}{2}\underset{\sim}{a} + \frac{3}{2}\underset{\sim}{b}$

Question 15

The angle between the vectors $-\underset{\sim}{i}+\underset{\sim}{k}$ and $2\underset{\sim}{i}+2\underset{\sim}{j}-\underset{\sim}{k}$ is

A. $\dfrac{\pi}{6}$

B. $\dfrac{\pi}{4}$

C. $\dfrac{3\pi}{4}$

D. $\dfrac{5\pi}{6}$

Question 16

The displacements, in metres, of two particles, A and B, moving in the x-y plane relative to an origin O at time t seconds are given by

$$\underset{\sim}{r}_{\mathrm{A}}(t)=\left(t^2-2t\right)\underset{\sim}{i}+(t-3)\underset{\sim}{j} \quad \text{and} \quad \underset{\sim}{r}_{\mathrm{B}}(t)=(1-t)\underset{\sim}{i}+(t-2)\underset{\sim}{j},$$

where $t \geq 0$.

The paths of these particles

A. intersect at $(3,-4)$ and $(0,-1)$.

B. intersect at $(3,-4)$ only.

C. intersect at $(0,-1)$ only.

D. do not intersect.

Question 17

The distance from the point $P(1,0,-2)$ to the plane $x-2y+2z=1$ is

A. $\dfrac{1}{3}$

B. $\dfrac{4}{3}$

C. 3

D. 4

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Do not write in this area.

Question 18

The area of the triangle with vertices $A(-6,0,2)$, $B(-4,2,0)$ and $C(2,-2,4)$ is

A. $2\sqrt{14}$

B. $2\sqrt{21}$

C. $6\sqrt{6}$

D. $10\sqrt{2}$

Question 19

Let X and Y be independent random variables such that $\text{sd}(X) = 1$ and $\text{sd}(Y) = 2$.

The standard deviation of $3X - 2Y$ is equal to

A. 5

B. 7

C. $\sqrt{11}$

D. $\sqrt{17}$

Question 20

The battery lives of smartphones is known to vary normally with a mean of μ hours and a standard deviation of 2 hours. From a random sample of smartphones, an approximate 90% confidence interval for μ was determined to be $(18.5468, 18.8532)$.

The size of the sample from which this confidence interval was obtained is

A. 21

B. 26

C. 461

D. 655

End of Section A

Section B

Instructions

- Answer **all** questions in the spaces provided.
- Write your responses in English.
- Unless otherwise specified, an **exact** answer is required to a question.
- In questions where more than one mark is available, appropriate working **must** be shown.
- Unless otherwise indicated, the diagrams in this book are **not** drawn to scale.
- Take the **acceleration due to gravity** to have magnitude g m s^{-2}, where $g = 9.8$

Question 1 (9 marks)

Consider the function $f(x) = \dfrac{x^3 - 2}{2x}$.

a. Find the coordinates of the stationary point of the graph of f. 1 mark

b. Find the coordinates of the point of inflection of the graph of f. 2 marks

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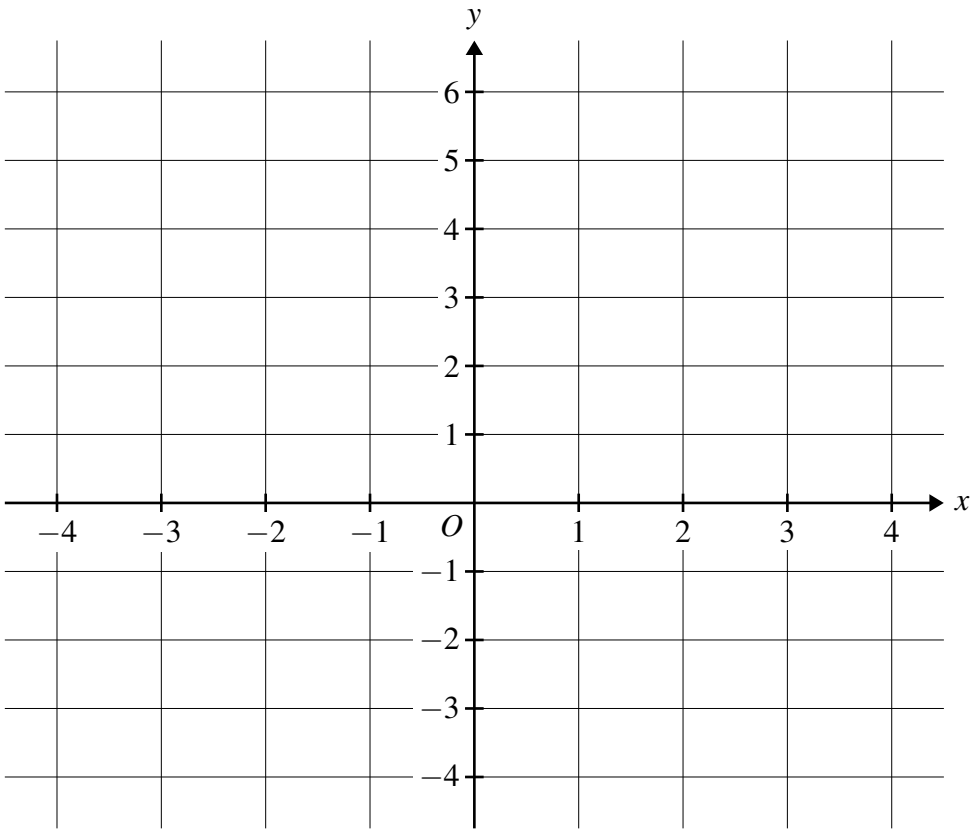
c. Sketch the graph of f on the axes below. Label the stationary point and the point of inflection with their coordinates and label the vertical asymptote with its equation. 3 marks



d. The region enclosed by the graph of f, the x-axis, and the lines $x = -2$ and $x = -1$ is rotated about the x-axis to form a solid of revolution.

Find the volume of this solid. 2 marks

Consider the function $g(x) = \dfrac{x^3 - 2}{2x + k}$, where k is a real constant.

e. Find the value of k for which the graph of g has no vertical asymptotes. 1 mark

Do not write in this area.

Question 2 (10 marks)

Let $u = -1 + \sqrt{3}i$.

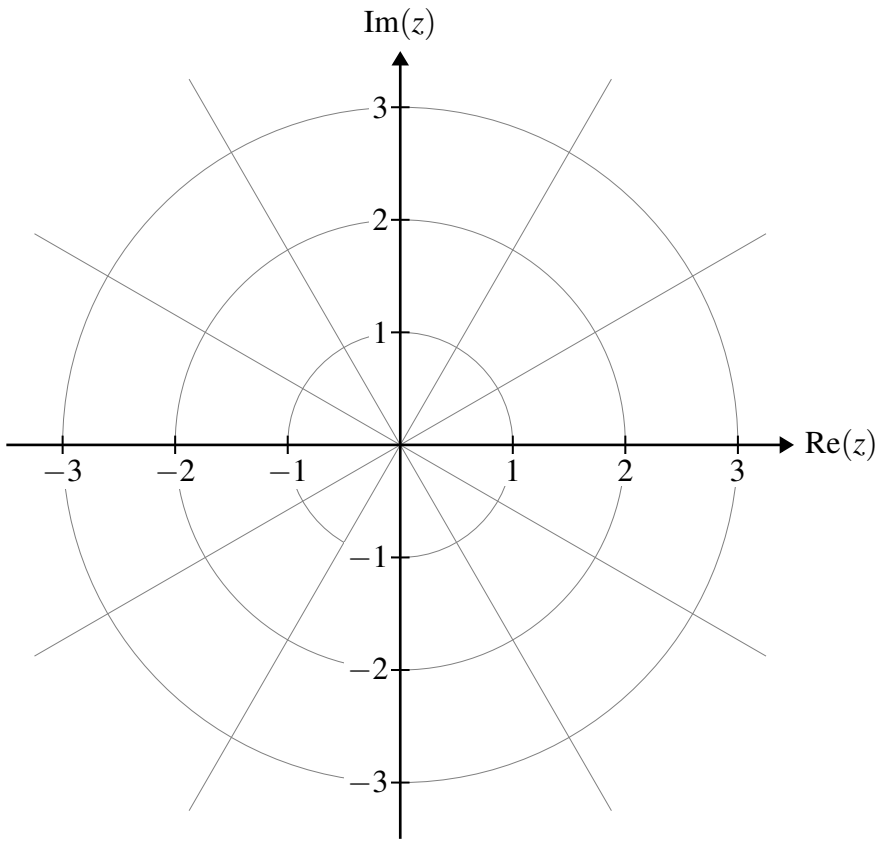
a. Express u in polar form. 1 mark

Consider the line defined by $L = \{z \in \mathbb{C} : |z| = |z - u|\}$.

b. Show that the relation $|z| = |z - u|$ can be expressed as $y = \dfrac{x}{\sqrt{3}} + \dfrac{2}{\sqrt{3}}$, where $z = x + yi$ and $x, y \in \mathbb{R}$. 2 marks

Consider the unit circle defined by $S = \{z \in \mathbb{C} : |z| = 1\}$.

c. On the Argand diagram below, sketch and label both L and S. 2 marks



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d. Consider the upper half plane defined by $H = \{z \in \mathbb{C} : \text{Im}(z) > 0\}$.

Find the values of $w \in \mathbb{C}$ and $\theta \in (-\pi, \pi]$ such that $\{z \in \mathbb{C} : \text{Arg}(z - w) = \theta\} = L \cap H$. 2 marks

e. **i.** On the Argand diagram in **part c**, shade the region defined by

$$\left\{z \in \mathbb{C} : 1 \leq |z| \leq |z - u|\right\} \cap \left\{z \in \mathbb{C} : \frac{2\pi}{3} \leq \text{Arg}(z) \leq \pi\right\}.$$ 1 mark

ii. Find the area of this region. 2 marks

Do not write in this area.

Question 3 (11 marks)

Consider the Gompertz differential equation

$$\frac{dW}{dt} = rW\left[1 - \log_e\left(\frac{W}{K}\right)\right], \quad W > 0,$$

which has been used as a model for tumour growth in mice. In this context, W is the weight of the tumour at time t, and r and K are positive constants.

a. Show that by making the substitutions $w = \dfrac{W}{K}$ and $\tau = rt$, the differential equation becomes $\dfrac{dw}{d\tau} = w\big(1 - \log_e(w)\big)$, for scaled weight w at scaled time τ. 2 marks

Consider a tumour with an initial scaled weight of $w(0) = e^{-1}$.

b. Solve the scaled differential equation $\dfrac{dw}{d\tau} = w\big(1 - \log_e(w)\big)$ with $w(0) = e^{-1}$ to determine the scaled weight $w(\tau)$ of the tumour for all $\tau \geq 0$. 3 marks

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c. Determine the limiting behaviour of $w(\tau)$ as $\tau \to \infty$. 1 mark

d. **i.** Using implicit differentiation, show that $\dfrac{d^2w}{d\tau^2} = w\log_e(w)\big(\log_e(w) - 1\big)$. 2 marks

ii. Hence, find the rescaled weight w of the tumour at which its rate of growth is maximum. 1 mark

If a continuous infusion therapy is introduced at $\tau = 0$, the differential equation modelling the scaled weight of a tumour can be revised to be

$$\frac{dw}{d\tau} = w\big(1 - \log_e(w)\big) - \lambda w,$$

where the positive constant λ determines the therapy-induced tumour death rate.

e. Find the values of λ for which the scaled weight of a tumour is guaranteed to never exceed e^{-1} for all initial conditions $w(0) \in \big(0, e^{-1}\big]$. 2 marks

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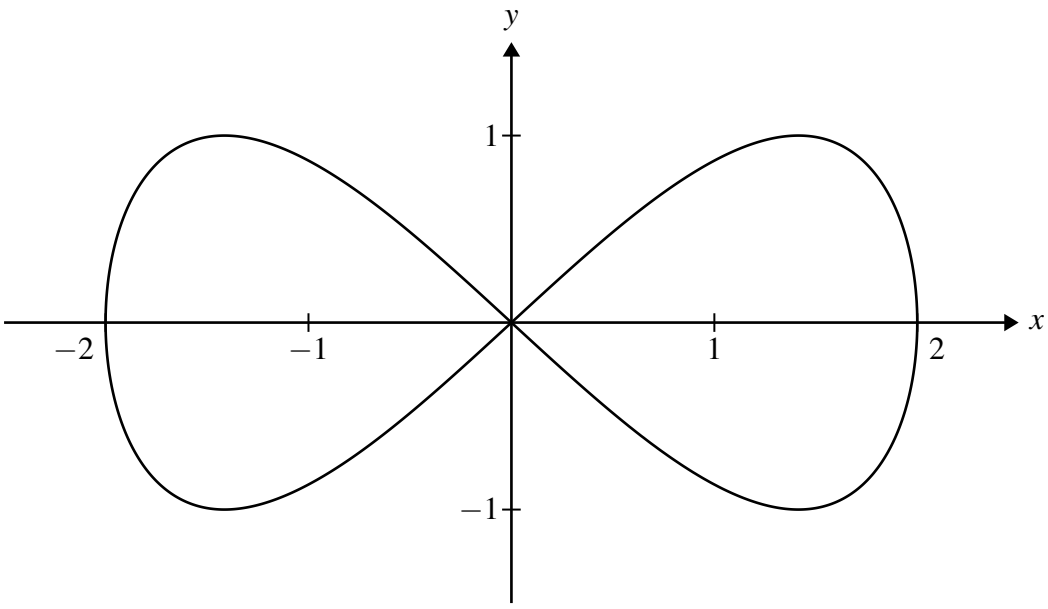
Question 4 (9 marks)

The displacement, in metres, of a particle moving in the x-y plane relative to an origin O at time t seconds is given by

$$\underset{\sim}{r}(t) = 2\cos\left(\frac{t}{2}\right)\underset{\sim}{i} + \sin(t)\underset{\sim}{j}, \quad t \geq 0.$$

a. Using a trigonometric identity, find the Cartesian equation of the path of the particle. Express your answer in the form $y^2 = f(x)$. 2 marks

b. On the diagram below, label the particle's initial position with its coordinates and use arrows to show its direction of motion along the path. 1 mark



c. Find the time taken, in seconds, for the particle to first return to its initial position. 1 mark

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d. Find the total distance travelled by the particle by the time it first returns to its initial position. Give your answer in metres, correct to two decimal places. 2 marks

e. Find the minimum distance, in metres, of the the particle from the point $(-1,0)$ and the first time, in seconds, at which this minimum occurs. Give your answers correct to one decimal place. 3 marks

Do not write in this area.

Question 5 (10 marks)

Let L_1 be the line in $\mathbb{R}^3$ with vector equation

$$\underset{\sim}{r}_1(t) = \underset{\sim}{i} - \underset{\sim}{j} + \underset{\sim}{k} + t\left(\underset{\sim}{i} - 2\underset{\sim}{j}\right), \quad t \in \mathbb{R}.$$

a. Find the coordinates of the point where L_1 intersects the x-z plane. 2 marks

Let L_2 be another line in $\mathbb{R}^3$ with Cartesian equations

$$-\frac{x}{3} = y = \frac{z-1}{3}.$$

b. Show that a vector equation of L_2 is given by

$$\underset{\sim}{r}_2(s) = \underset{\sim}{k} + s\left(-3\underset{\sim}{i} + \underset{\sim}{j} + 3\underset{\sim}{k}\right), \quad s \in \mathbb{R}.$$ 1 mark

c. Show that the lines L_1 and L_2 are skew. 2 marks

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d. Find the distance between L_1 and L_2. 2 marks

Let P_1 and P_2 be the parallel planes such that P_1 contains L_1 and P_2 contains L_2.

e. Find the Cartesian equations of P_1 and P_2. 3 marks

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Question 6 (11 marks)

Road safety cameras are commonly used to monitor and enforce vehicle speed limits.

On a road with a speed limit of 40 km h^{-1}, one such device measures the speed of each passing vehicle three times. Each measurement is independent and follows a normal distribution with a mean of μ km h^{-1} and a standard deviation of 2 km h^{-1}, where μ represents the true speed of the vehicle.

Traffic police perform a statistical test at the 1% level of significance using the average of the three measurements to determine whether a vehicle's true speed exceeds the speed limit of 40 km h^{-1}. If there is sufficient evidence of a speeding offence, a fine is issued to the driver.

a. Write down suitable null and alternative hypotheses for this test. 1 mark

b. A van travelling on this road has the following recorded speeds:

$$42.2 \text{ km h}^{-1}, \quad 43.4 \text{ km h}^{-1}, \quad 41.9 \text{ km h}^{-1}.$$

i. Calculate the p value for the statistical test, correct to four decimal places. 2 marks

ii. Determine, with justification, whether traffic police should issue a speeding fine to the driver based on these measurements. 1 mark

c. Find the critical sample mean for this test. Give your answer in kilometres per hour, correct to two decimal places. 1 mark

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d. **i.** Give an interpretation of a type II error in this context. 1 mark

ii. Find the true speed of a vehicle travelling on this road for which the probability of a type II error occurring is 0.1. Give your answer in kilometres per hour, correct to one decimal place. 2 marks

e. A car travelling on this road has a true speed of 43 km h^{-1}.

Given that one of the three recorded speeds of the car is is 44.6 km h^{-1}, find the probability that the driver will be fined by traffic police. Give your answer correct to three decimal places. 3 marks

End of Question and Answer Book

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